



Sydney Girls High School

2016

YEAR 11 Preliminary Course

ASSESSMENT TASK 3

MATHEMATICS

Time Allowed: 60 minutes + 5 minutes reading time

Total: 60 marks

Topics: Basic Arithmetic and Algebra, Plane Geometry, Real Functions, Trigonometric Ratios, The Linear Function, Introductory Calculus: Tangent to a curve, differentiation of a function, The Quadratic Polynomial and Locus.

General Instructions:

- There are Four (4) Questions which are of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each Question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.

Student Name : _____

Teacher Name : _____

QUESTION 1**Marks**

(a) Evaluate $255(0.0024)^2$ giving your answer correct to 3 significant figures. 2

(b) Solve $x^2 + 9x - 10 = 0$. 2

(c) Solve $|3 - 2x| > 5$. 2

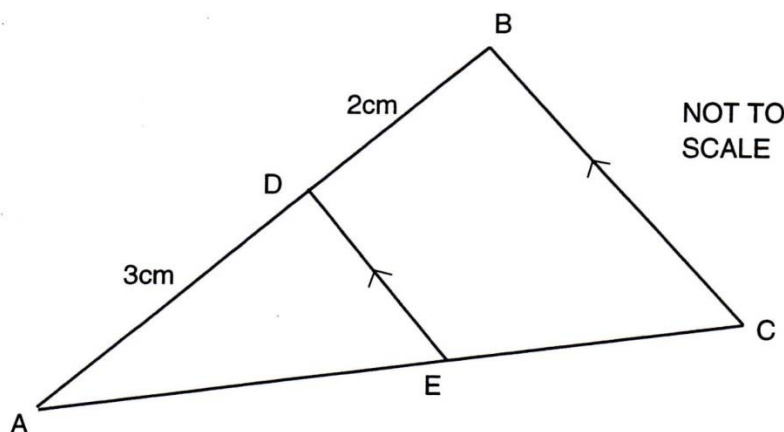
(d) Factorise $p^3 + 8$. 1

(e) Solve $\sin \beta = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \beta \leq 360^\circ$. 2

(f) Express $\frac{9}{5 - \sqrt{7}}$ in the form $a + b\sqrt{7}$, where a and b are rational. 2

(g) Simplify $\sqrt{(a - 5)(a + 5) + 25}$ where $a > 0$. 2

(h)



Copy or trace the diagram onto your worksheet.

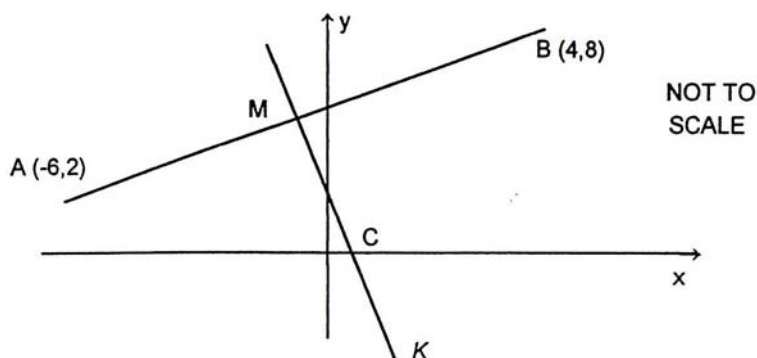
In $\triangle ABC$, points D and E lie on lines AB and AC respectively, such that DE is parallel to BC . If AD is 3 centimetres and DB is 2 centimetres, find the ratio $DE:BC$, giving reasons. 2

QUESTION 2 (Start a new page)**Marks**

(a) If $f(x) = 3x^4 - 5$, find $f'(-3)$.

2

(b)



In the diagram above, A is the point $(-6, 2)$ and B $(4, 8)$.

(i) Find the point M, the midpoint of AB.

1

(ii) Find the gradient of the line AB.

1

(iii) Line K is the perpendicular bisector of line AB. Find the equation of line K.

2

(iv) Find the coordinates of C, the point at which line K cuts the x-axis.

1

(c) Find the equation of the normal to the curve $y = \frac{3}{x} + x$ at the point where $x = 2$.

2

(d) Find the value(s) of k for which $x^2 + (k - 1)x + 4 = 0$ has equal roots.

2

(e) State the domain and range of the function $y = \sqrt{36 - x^2}$.

2

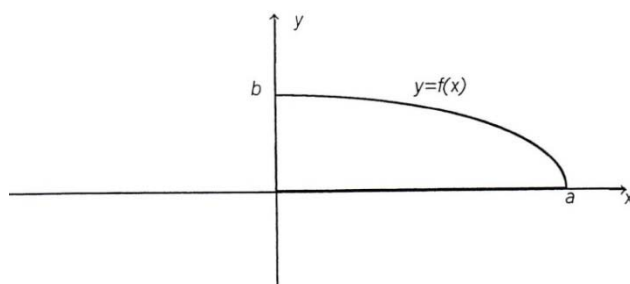
(f) Draw a neat sketch showing the region on the number plane where the inequalities

2

$x^2 + y^2 \leq 4$ and $y \geq x$ hold simultaneously.

QUESTION 3 (Start a new page)**Marks**

- (a) The diagram below shows part of the graph of the function
- $y = f(x)$
- .

1Copy the diagram and complete the graph if $y = f(x)$ is an odd function.

- (b) Form a quadratic equation with roots
- $(1 + \sqrt{3})$
- and
- $(1 - \sqrt{3})$
- .

2

- (c) Differentiate each of the following:

(i) $\frac{3x}{x^2 - 2}$.

2

(ii) $(2 - 3x)^7$.

1

- (d) Consider the parabola
- $y = \frac{1}{2}x^2 - x + \frac{3}{2}$
- .

(i) Find the coordinates of the vertex.

1

(ii) Find the coordinates of the focus

1

(iii) Find the equation of the directrix.

1

- (e) Given that
- α
- and
- β
- are the roots of the quadratic equation
- $3x^2 - 5x - 1 = 0$
- , find the values of:

(i) $\frac{1}{\alpha} + \frac{1}{\beta}$.

2

(ii) $(\alpha^2 + \beta^2)^2$

2

- (f) Prove the following identity:

2

$$\sec \theta + \tan \theta = \frac{\cos \theta}{1 - \sin \theta}.$$

QUESTION 4 (Start a new page)**Marks**

(a) If $\tan \beta = -2$ and $0^\circ < \beta < 270^\circ$, find the value of $\operatorname{cosec} \beta$ in surd form. 2

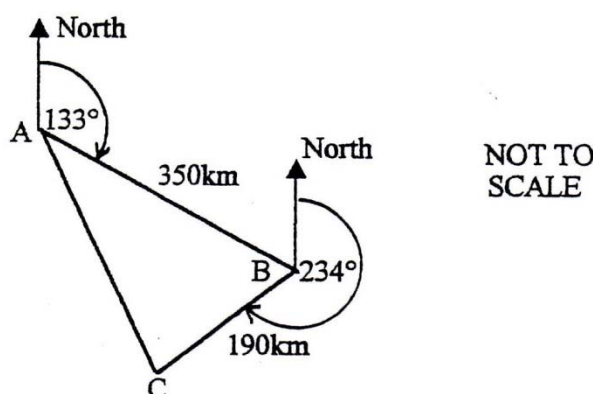
(b) The equation of a locus is given by $x^2 + y^2 - 2x + 4y - 11 = 0$.

(i) Show that this is the equation of a circle and that the centre is the point $(1, -2)$. 2

(ii) Find the length of its radius. 1

(iii) Does the line $2x - y + 8 = 0$ intersect the circle? Explain. 2

(c) A plane leaves airport A and flies on a bearing of 133° a distance of 350 kilometres to point B. It then flies on a bearing of 234° for 190 kilometres to point C, as shown in the diagram.



Copy the diagram onto your worksheet.

(i) Show that $\angle ABC = 79^\circ$. 1

(ii) Find the distance from C to A, to the nearest kilometre. 2

(iii) Find the bearing of A from C, to the nearest degree. 2

(d) The tangent to $y = ax + \frac{b}{x^2}$ at $(2, 0)$ makes an angle of 60° with the positive direction of the x-axis. Find the values of a and b . 3

END OF PAPER

Question 1

$$\begin{aligned}
 \text{a) } 255(0.0024)^2 &\doteq 0.0014688 \dots \\
 &= 1.47 \times 10^{-3} \text{ (3 sig figs)} \\
 &\text{or } 0.00147 \text{ (3 sig figs)}
 \end{aligned}$$

* one mark awarded for the correct answer, 1 mark for converting correctly to 3 sig figs.

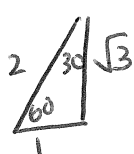
$$\begin{aligned}
 \text{b) } x^2 + 9x - 10 &= 0 \\
 (x+10)(x-1) &= 0 \\
 \therefore x+10 &= 0 \quad x-1 = 0 \\
 x &= -10 \quad x = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } |3-2x| &> 5 \\
 3-2x &> 5 \quad 3-2x < -5 \\
 -2x &> 2 \quad -2x < -8 \\
 x &< -1 \quad x > 4
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } p^3 + 8 &= p^3 + 2^3 \\
 &= (p+2)(p^2 - 2p + 4)
 \end{aligned}$$

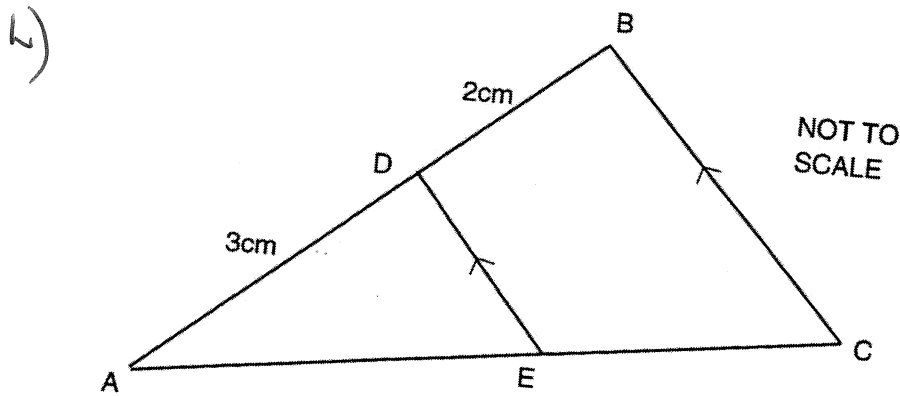
$$\text{e) } \sin \beta = \frac{\sqrt{3}}{2}$$

$$\begin{aligned}
 \beta &= 60^\circ, 180-60^\circ \\
 &= 60^\circ, 120^\circ
 \end{aligned}$$



$$\begin{aligned}
 \text{f) } \frac{9}{5-\sqrt{7}} \times \frac{5+\sqrt{7}}{5+\sqrt{7}} &= \frac{45+9\sqrt{7}}{25-7} \\
 &= \frac{9(5+\sqrt{7})}{18} \\
 &= \frac{5}{2} + \frac{\sqrt{7}}{2}
 \end{aligned}$$

$$\begin{aligned}
 9) & \sqrt{(a-5)(a+5)+25} \\
 &= \sqrt{a^2-25+25} \\
 &= \sqrt{a^2} \\
 &= a, a > 0
 \end{aligned}$$



In $\triangle ABC$ and $\triangle ADE$

$\angle A$ is common

$\angle ADE = \angle ABC$ (corresponding $\angle$ s $DE \parallel BC$)

$\therefore \triangle ABC \sim \triangle ADE$ (equiangular)

$\therefore \frac{DE}{BC} = \frac{AD}{AB}$ (matching sides similar $\triangle$ s in same proportion)

$$\frac{DE}{BC} = \frac{3}{5}$$

$$DE:BC = 3:5$$

Question 2 (15 Marks)

a) $f(x) = 3x^4 - 5$

$$f'(x) = 12x^3$$

$$\therefore f'(-3) = 12(-3)^3$$

$$= 12 \times -27$$

$$= \underline{\underline{-324}} \quad (2)$$

b) i) $M = \left(\frac{-6+4}{2}, \frac{2+8}{2} \right) = \underline{\underline{(-1, 5)}} \quad (1)$

ii) $m_{AB} = \frac{8-2}{4+6} = \underline{\underline{\frac{3}{5}}} \quad (1)$

iii) $m_K = -\frac{5}{3}$ and $M = (-1, 5)$

$\therefore$ Equation of line k:

$$y - 5 = -\frac{5}{3}(x + 1)$$

$$y = -\frac{5}{3}x - \frac{5}{3} + 5$$

(2)

$$\underline{\underline{y = -\frac{5}{3}x + \frac{10}{3}}} \quad \text{OR} \quad \underline{\underline{5x + 3y - 10 = 0}}$$

iv) When $y = 0$: $\frac{5}{3}x = \frac{10}{3} \quad \therefore \underline{\underline{C = (2, 0)}} \quad (1)$

$$x = 2$$

* Many students substituted $x=0$ (instead of $y=0$) to find the x -intercept.
* A few students did not give their answer as a point.

c) $y = 3x^{-1} + x$

$$\frac{dy}{dx} = -3x^{-2} + 1$$

at $x=2$: $m = -\frac{3}{2^2} + 1 = \frac{1}{4}$; $y = \frac{7}{2}$

$\therefore$ Equation of normal:

$$y - \frac{7}{2} = -4(x - 2)$$

$$\underline{\underline{y = -4x + \frac{23}{2}}} \quad \text{OR} \quad \underline{\underline{8x + 2y - 23 = 0.}} \quad (2)$$

d) $x^2 + (k-1)x + 4 = 0.$

For equal roots: $\Delta = 0$

$$b^2 - 4ac = 0$$

$$(k-1)^2 - 4(1)(4) = 0$$

$$(k-1)^2 - 16 = 0$$

$$(k-1)^2 = 16$$

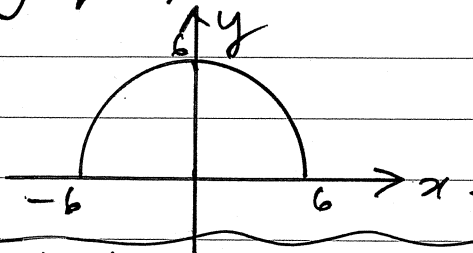
$$k-1 = \pm 4$$

$$\therefore \underline{\underline{k = 5 \text{ or } -3.}} \quad (2)$$

e) $y = \sqrt{36 - x^2} \rightarrow$ graph of semi-circle

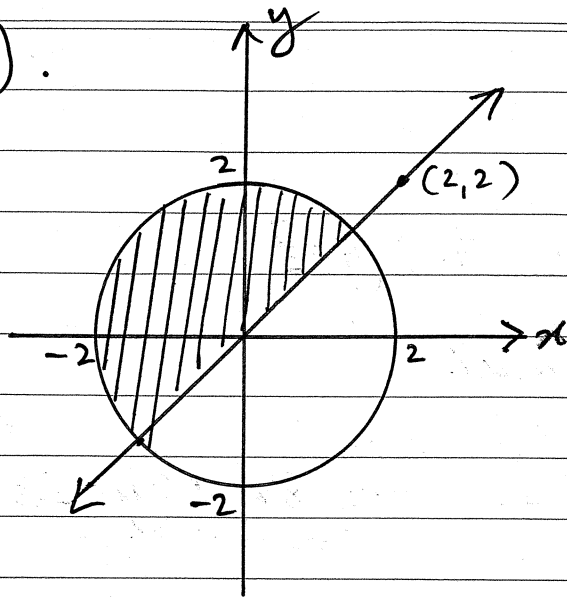
D: $-6 \leq x \leq 6$

R: $0 \leq y \leq 6$



★ Many students lost marks with the range being incorrectly stated

f).



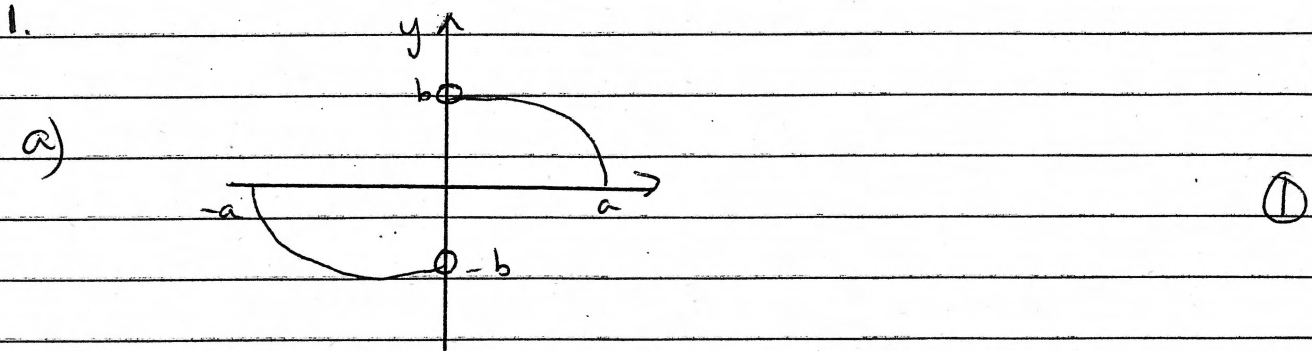
2.

region ✓
graphs ✓

* Many students were unsure whether to dot the lines or not. Solid lines are drawn with equality.

* Diagrams need to be drawn in PEN

Question 3



b) $x^2 - (1+\sqrt{3} + 1-\sqrt{3})x + (1+\sqrt{3})(1-\sqrt{3})$
 $x^2 - 2x - 2 = 0$ ②

c) $\frac{d}{dx} \left(\frac{3x}{x^2-2} \right) = \frac{(x^2-2)(3) - 3x(2x)}{(x^2-2)^2}$
 $= \frac{3x^2 - 6 - 6x^2}{(x^2-2)^2}$ ②
 $= \frac{-3x^2 - 6}{(x^2-2)^2} \Rightarrow \frac{-3(x^2-2)}{(x^2-2)^2}$
 $= \frac{-3}{x^2-2}$

ii) $\frac{d}{dx} (2-3x)^7 = 7(2-3x)^6(-3)$
 $= -21(2-3x)^6$ ①

d) $y = \frac{1}{2}x^2 - x + \frac{3}{2}$
 $2y = x^2 - 2x + 3$
 $x^2 - 2x + 3 = 2y$
 $x^2 - 2x + 1 = 2y - 2$
 $(x-1)^2 = 2(y-1)$

∴ i) $V(1,1)$ ✓ ii) $4a = 2 \Rightarrow a = \frac{1}{2}$ ∴ $Fat(1, \frac{3}{2})$ ✓
 directrix $y = \frac{1}{2}$ ✓
 1 CFPA 1 1 ③
 applies here.

$$e) \quad 1) \quad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha \beta} \quad , \quad \alpha + \beta = \frac{5}{3}, \quad \alpha \beta = -\frac{1}{3}$$

$$= \frac{5}{3} \div -\frac{1}{3}$$

$$= -5 \quad (2)$$

$$11) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(\frac{5}{3}\right)^2 - 2\left(-\frac{1}{3}\right)$$

$$= \frac{31}{9}$$

$$\therefore (\alpha^2 + \beta^2)^2 = \frac{961}{81} \quad (11 \frac{70}{81}) \quad (2)$$

$$f) \quad \sec \theta + \tan \theta$$

$$= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 + \sin \theta}{\cos \theta} \times \frac{1 - \sin \theta}{1 - \sin \theta}$$

$$= \frac{1 - \sin^2 \theta}{\cos \theta (1 - \sin \theta)}$$

$$= \frac{\cos^2 \theta}{\cos \theta (1 - \sin \theta)}$$

$$= \frac{\cos \theta}{1 - \sin \theta} \quad (2)$$

$$\text{OR} \quad \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$$

$$= \frac{\cos \theta + \cos \theta \sin \theta}{\cos^2 \theta}$$

$$= \frac{\cos \theta (1 + \sin \theta)}{1 - \sin^2 \theta}$$

$$= \frac{\cos \theta (1 + \sin \theta)}{(1 + \sin \theta)(1 - \sin \theta)}$$

$$= \frac{\cos \theta}{1 - \sin \theta}$$

4

Note: Most students got it right, however, some were confused about the use of quadrants.

a)

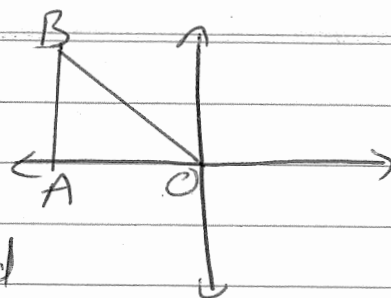
$$\tan \beta = -2$$

$$0^\circ < \beta < 360^\circ$$

$\tan$ is -ve in II and IV

quadrant, as β is restricted to be in the range 0° to 360° , so it is in II quadrant

$$\tan \beta = \frac{AB}{OA} = -2$$



$$\Rightarrow AB = 2, OA = -1, OB = \sqrt{4+1} = \sqrt{5}$$

$$\operatorname{cosec} \beta = \frac{OB}{AB} = \frac{\sqrt{5}}{2}$$

Alternative $\tan \beta = -2$

$$\beta = 116^\circ 33' 54''$$

$$\operatorname{cosec} \beta = \frac{1}{\sin \beta} = \frac{\sqrt{5}}{2}$$

b)

$$x^2 + y^2 - 2x + 4y - 11 = 0$$

some were confused about the process of completing the square

$$x^2 - 2x + 1 - 1 + y^2 + 4y + 4 - 11 - 4 = 0$$

$$(x-1)^2 + (y+2)^2 = 16$$

centre $(1, -2)$ radius = 4

(ii)

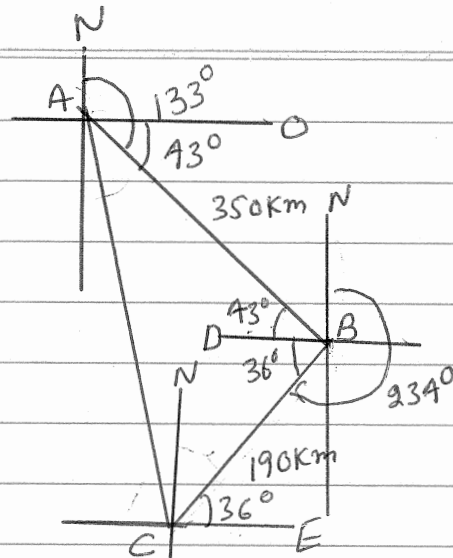
$$\perp \text{ distance of the centre of the circle from the given line} = \frac{|2+2+8|}{\sqrt{4+1}} = \frac{12}{\sqrt{5}}$$

$$\text{As } \frac{12}{\sqrt{5}} > 4$$

Hence $\perp$ distance is greater than the radius of the circle, so the given line does not touch the circle.

Different approaches, however, most of the students got it right.

c)



i. $\angle OAB = \angle ABD =$ alt. angles

$$\angle DBC = 36^\circ$$

$$\angle ABC = \angle ABD + \angle DBC = 43^\circ + 36^\circ = 79^\circ$$

(ii) using cosine rule

$$AC^2 = AB^2 + BC^2 - 2AB \cdot BC \cos 79^\circ$$

$$AC^2 = 350^2 + 190^2 - 2 \times 350 \times 190 \cos 79^\circ$$

$$AC \approx 365 \text{ km}$$

(iii) $\angle BCE = 36^\circ$

$$\angle NCB = 90^\circ - 36^\circ = 54^\circ$$

using sine rule

$$\frac{\sin 79^\circ}{AC} = \frac{\sin C}{350}$$

$$\sin C = \frac{350 \times \sin 79^\circ}{AC}$$

$$C = \sin^{-1} \left(\frac{350 \times \sin 79^\circ}{365} \right) = 70^\circ 16' 8''$$

$$\angle NCA = 70^\circ 16' 8'' - 54^\circ = 16^\circ 16' 8''$$

$$\text{Bearing of A from C} = 360^\circ - 16^\circ 16' 8''$$

$$= 343^\circ 43' 52''$$

$$\approx 344^\circ$$

Majority got it correct.

d)

$$y = ax + \frac{b}{x^2}$$

$$\text{gradient of tangent } m = \frac{dy}{dx} = a - 2bx^{-3}$$

$$m = a - \frac{2b}{x^3}$$

$$\text{Also } m = \tan 60^\circ$$

$$\Rightarrow a - \frac{2b}{x^3} = \sqrt{3}$$

$$\text{point } (2, 0)$$

$$a - \frac{2b}{8} = \sqrt{3} \quad - (1)$$

As $(2, 0)$ lies on the curve

$$2a + \frac{b}{4} = 0$$

$$8a + b = 0 \quad - (2)$$

From (1)

$$8a - 2b = 8\sqrt{3} \quad - (3)$$

$$(3) - (2)$$

$$-3b = +8\sqrt{3}$$

$$b = -\frac{8}{\sqrt{3}}$$

Sub in (2)

$$8a + \frac{-8}{\sqrt{3}} = 0$$

$$a - \frac{1}{\sqrt{3}} = 0$$

$$a = \frac{1}{\sqrt{3}}$$

Some students made the process of solving of simultaneous equations very complicated. Some did not simplify their answers. Few had done differentiation wrong.